

Expense Tracker – Project Documentation

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1. Technology Stack

The application is built using plain web technologies with no installation required. The chosen stack is HTML, CSS, and JavaScript. This combination was selected because it runs directly in any modern browser without needing a server, a build tool, or any additional setup.

HTML5 is used for the structure and layout of the application. Semantic elements like header, form, and section tags are used where appropriate.

CSS3 handles all the visual styling. Flexbox and Grid are used for layout. The design is responsive and adapts to different screen sizes using media queries.

Vanilla JavaScript (ES6+) contains all the application logic. No external frameworks or libraries are used for the core functionality. This keeps the codebase simple and easy to understand.

LocalStorage API is used to persist data in the browser. When the user closes and reopens the app, all expenses are still there. Data is stored as a JSON string under a single key.

Chart.js (loaded via CDN) is used for the visual charts in the monthly summary section. It provides a doughnut chart and a bar chart with minimal configuration.

No backend, no database, no server-side code. Everything runs on the client side.

2. Application Architecture

The application follows a simple single-page architecture. There is one HTML file that defines the layout, one CSS file for styles, and one JavaScript file that handles all the logic.

File Structure

```
expense-tracker/  
├── index.html    - page structure and UI components  
├── style.css     - all styles and responsive rules  
└── app.js       - data layer, rendering, event handling
```

Data Flow

User interactions trigger JavaScript functions which read from or write to LocalStorage. After any data change, the relevant parts of the DOM are re-rendered to reflect the current state.

User Action → app.js function → LocalStorage (read/write) → DOM update

Data Schema

Expenses are stored in LocalStorage as a JSON array under the key `expenses_data`. Each expense object has the following structure:

```
{
  "id": "unique string (timestamp + random)",
  "title": "Grocery shopping",
  "amount": 45.50,
  "category": "Food",
  "date": "2026-06-03",
  "note": "weekly groceries"
}
```

The `id` field is generated from the current timestamp combined with a random string. This ensures each expense has a unique identifier without needing a database sequence.

3. Application Functionality

Adding an Expense

The user clicks the "Add Expense" button in the top right corner. A modal dialog appears with a form. The user fills in the title, amount, category, date, and an optional note. Clicking Save stores the new expense in LocalStorage and updates the list.

Editing an Expense

Each expense card in the list has an Edit button. Clicking it opens the same modal form pre-filled with the existing data. The user can modify any field and click Save to update. The original expense is identified by its `id`.

Deleting an Expense

Each expense card also has a Delete button. Clicking it shows a confirmation dialog. If confirmed, the expense is removed from LocalStorage and the list is re-rendered.

Filtering

The filter bar at the top allows filtering the expense list by category, by month, and by year. Filters can be combined. The total amount shown above the list updates to reflect only the filtered expenses.

Monthly Summary

The lower section of the page shows a monthly summary for a selected month and year. It includes

a doughnut chart (spending by category), a bar chart (same data in bar form), and a set of stat cards showing total spent, number of transactions, average per expense, and the category with the highest spending.

4. User Interface Description

The layout is divided into three main areas:

Header – contains the app title on the left and the "Add Expense" button on the right.

Expense Section – contains the filter controls, a total bar showing the filtered total and count, and the list of expense cards. Each card shows a colored dot for the category, the title, a category badge, the date, an optional note, the amount, and Edit/Delete buttons.

Monthly Summary Section – contains two dropdowns to select the month and year, two charts side by side (doughnut and bar), and stat cards below the charts.

Modal – a centered overlay dialog for adding and editing expenses. It includes form fields for title, amount, category, date, and note, plus Cancel and Save buttons.

The color scheme is calm and neutral with blue as the primary action color. Each category has a distinct color used consistently across the category dot, the category badge, and the chart slices.

The interface is responsive and works on smaller screens. On mobile, the charts stack vertically and the filter controls wrap to new lines.

5. Summary

This project is a single-page web application for tracking personal expenses. It is built with HTML, CSS, and JavaScript and stores all data in the browser's LocalStorage. No installation or server is needed — the user just opens the HTML file in a browser.

The main features are adding, editing, and deleting expenses; filtering the list by category, month, and year; and viewing a monthly summary with charts and statistics.

The application satisfies all the requirements in the project specification: it provides core functionality without authentication, works as a single-user app, and uses a flexible single-layer architecture.